

Grounded Ontology Walker: Interpretable Diagnostic Reasoning and Report Generation for Whole Slide Images

Anonymous

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Abstract. Generating a pathology report from a whole-slide image (WSI) is increasingly framed as a reasoning problem: a model should not only produce the report but also expose the diagnostic reasoning behind it. Free-form generated reasoning is difficult to verify and can diverge from the reported findings. We present the Grounded Ontology Walker (GOW), a discriminative pipeline in which the reasoning is an explicit walk over a diagnostic ontology mined from expert reasoning chains, and the report is a deterministic render of that walk. Every reasoning step is a valid ontology transition, so the chain is auditable, and because the report is derived from the chain it is faithful to the reasoning by construction. Each diagnostic question is answered from the specific image regions that a question-conditioned selector attends to, over patch features from Virchow2, a frozen pathology foundation model, which grounds and localizes every decision. To remain safe on organs that are absent from training, GOW includes a per-organ distribution-shift detector that flags unseen organs and routes them through the nearest in-ontology topology with open-vocabulary naming, and we show it generalizes to novel organs on public external slides. On the blind Test Phase 1 evaluation GOW attains an overall score of 0.791: visual grounding is perfect (1.00, spanning background rejection, input sensitivity, and cross-region consistency), the reasoning graph is recovered at edge F1 0.852 with matched-edge answer similarity 0.772, and the report scores 0.572. On the development split held out from the training data, reasoning-edge F1 is 0.922 and organ recognition 0.993. We report performance per metric and evaluation setting and analyze the contribution of each component.

Keywords: Computational pathology · Report generation · Reasoning · Multiple instance learning · Out-of-distribution detection.

1 Introduction

Automated report generation from whole-slide images (WSIs) is increasingly a joint task: produce a structured pathology report together with the diagnostic reasoning that supports it, so the output is auditable. When the reasoning comes from a free-form generator, however, individual steps are not guaranteed valid and the rationale may be inconsistent with the reported findings.

This work targets the REG2026 (MICCAI 2026) challenge: given an H&E WSI, a system produces an explicit reasoning chain over a closed clinical question ontology plus a structured report (the workflow-reasoning interface) and answers grounding questions on regions of interest (the visual-grounding interface), across seven organs. Expert reasoning in the training data follows a bounded set of questions and transitions, defining a diagnostic ontology.

We formalize the reasoning as a traversal of this ontology, mined as a transition graph from the expert reasoning chains. Inference is a deterministic walk over the graph, and the report is rendered deterministically from the answers collected along the walk. We refer to the method as the Grounded Ontology Walker (GOW).

Contributions. (i) The chain of thought is a deterministic walk over a transition graph mined from expert chains, so every step is a valid, auditable ontology transition, not free-form text. (ii) The report is a deterministic render of the completed walk, faithful to the reasoning by construction. (iii) A question-conditioned pooler selects the patches that answer each question, scored against text embeddings of the allowed responses, supporting open-vocabulary answering of rare classes. (iv) A detector on the frozen features flags organs absent from training and routes them safely. The critical path is discriminative and LLM-free, hence reproducible and light enough for an offline single-GPU container.

2 Related Work

Report generation typically couples a WSI encoder to a text decoder, increasingly a medical language or multimodal model [14,16,11], fluent but with little guarantee the findings are grounded in tissue. **WSI foundation models** [22,1,19,15] give transferable patch features and slide-level variants [17,2] summarize a slide; we use one tile encoder as a frozen backbone and a vision-language model [9] as a text-embedding space for answers. **Attention-based MIL** [4,8] aggregates patches with an interpretable attention map; we use a gated variant with a question-conditioned pooler. **Structured-reasoning** methods encode clinical decisions as templates or graphs; we instead mine the decision graph from data and make inference a deterministic walk inside a closed, verifiable vocabulary.

3 Methods

Figure 1 shows the full pipeline.

3.1 Tiling, encoding, and the slide bag

Tissue is segmented with a quality-control model [18] and the slide is tiled into 224-pixel patches at 20x magnification, giving a variable-size bag of patches per slide. Each patch is embedded by the frozen Virchow2 [22] foundation model

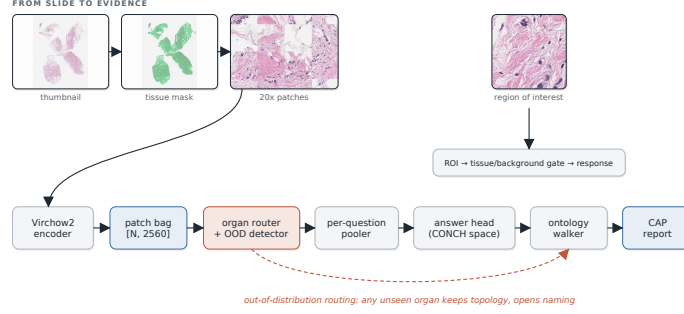


Fig. 1. The Grounded Ontology Walker. Top: input stages on one slide (thumbnail, GrandQC tissue mask, 20x patches, region of interest). Bottom: a frozen Virchow2 encoder embeds each patch into a bag; a gated-attention MIL head predicts the organ and feeds the distribution-shift (OOD) detector; reasoning then proceeds question by question, a question-conditioned pooler selecting patches, an answer head scoring them against text embeddings of the allowed answers, and a mined transition table supplying the next question, until the walk is rendered into a report. The dashed arrow routes an unseen organ through the nearest in-ontology topology.

into a 2560-dimensional vector formed by concatenating the class token with the mean of the patch tokens. Because the encoder is frozen, the bag is precomputed once per slide and reused by every downstream head. A slide thus yields a bag $H = \{\mathbf{h}_i\}_{i=1}^N$ with $\mathbf{h}_i \in \mathbb{R}^d$, $d=2560$.

3.2 Slide-level MIL: organ and global context

A gated-attention MIL head [4] aggregates the bag into a slide vector and predicts the organ. With $\phi_i = \text{ReLU}(\mathbf{W}\mathbf{h}_i + \mathbf{b}) \in \mathbb{R}^m$ ($m=512$), the attention weights and slide vector are

$$a_i = \frac{\exp(\mathbf{w}^\top (\tanh(\mathbf{V}\phi_i) \odot \sigma(\mathbf{U}\phi_i)))}{\sum_j \exp(\mathbf{w}^\top (\tanh(\mathbf{V}\phi_j) \odot \sigma(\mathbf{U}\phi_j)))}, \quad \mathbf{z} = \sum_i a_i \phi_i, \quad (1)$$

and the organ posterior is $\mathbf{p} = \text{softmax}(\mathbf{W}_o \mathbf{z})$; the gate lets a few informative patches dominate $\mathbf{z}$. The bag mean $\bar{\mathbf{h}} = \frac{1}{N} \sum_i \mathbf{h}_i$ is the input to the distribution-shift detector (Section 3.5), and $\mathbf{p}$ conditions the answer heads on organ context.

3.3 Per-question evidence selection

Reasoning proceeds one ontology question at a time. A question with CONCH text embedding $\mathbf{q}$ is projected into the patch space, $\tilde{\mathbf{q}} = \mathbf{W}_q \mathbf{q}$, and each patch receives a question-specific relevance $r_i = \text{MLP}([\mathbf{h}_i; \tilde{\mathbf{q}}])$. The evidence combines a soft-attention pool with a focal max-pool,

$$b_i = \text{softmax}_i(r_i), \quad \mathbf{e}_q = \rho \left(\underbrace{\sum_i b_i \mathbf{h}_i}_{\text{soft pool}} + \underbrace{\mathbf{h}_{\arg \max_i r_i}}_{\text{focal max}} \right). \quad (2)$$

The focal term matters clinically: a small but decisive focus (a microscopic tumor focus, a mitotic figure, a granuloma) can determine the answer despite occupying a tiny fraction of the slide. Because r_i is question-conditioned, different questions attend to different patches, and b_i localizes the tissue behind each decision.

3.4 Open-vocabulary answer heads

Answers are scored in the text space of CONCH [9], a pathology vision-language model whose image-text pretraining aligns morphology with terminology, so an answer is scored by the similarity of the evidence to its text description and rare or unseen classes stay reachable. Let $\mathcal{A}_q$ be the answers allowed at the current node, each with a CONCH text embedding $\mathbf{t}_a$. The evidence, slide vector, and organ posterior are projected to a query $\mathbf{g} = \pi([\mathbf{e}_q; \mathbf{z}; \mathbf{p}])$, and answers are scored contrastively in the CLIP sense [13],

$$s(a) = \tau \frac{\mathbf{g}^\top \mathbf{t}_a}{\|\mathbf{g}\| \|\mathbf{t}_a\|}, \quad \hat{a} = \arg \max_{a \in \mathcal{A}_q} s(a), \quad (3)$$

with a learned temperature τ . Restricting $\mathcal{A}_q$ to the node’s allowed answers gives organ- and context-masking for free and keeps every emitted string inside the ontology.

3.5 Distribution-shift detection and safe routing

The seven training organs are not an exhaustive list of the anatomy that can appear, so a slide may belong to an organ the seven-way router cannot represent. We standardize the frozen bag mean and project it to a $k=128$ subspace, $\mathbf{u} = \mathbf{P}(\bar{\mathbf{h}} - \boldsymbol{\mu})/\boldsymbol{\sigma}$. Rather than a single shared covariance, we estimate a per-organ mean $\boldsymbol{\nu}_c$ and a ridge-regularized per-organ covariance $\boldsymbol{\Sigma}_c$, and score a slide by its squared Mahalanobis distance to the nearest training organ [5],

$$D(\bar{\mathbf{h}}) = \min_c (\mathbf{u} - \boldsymbol{\nu}_c)^\top \boldsymbol{\Sigma}_c^{-1} (\mathbf{u} - \boldsymbol{\nu}_c). \quad (4)$$

The quadratic per-organ form is what lets the detector generalize to any of the seven routes rather than a single one: leave-one-organ-out analysis (Section 4) improves from 0.85 to 0.98 AUC when the shared covariance is replaced by per-organ covariances. A slide routed to organ c is flagged when $D(\bar{\mathbf{h}}) > \tau_c$, with τ_c fixed label-free from organ c ’s in-distribution distances and scaled for precision; a false flag on a seen organ costs more than a missed borderline novel organ, so only about 2% of held-out in-distribution slides are flagged. A flagged slide walks the same routed topology, named by open-vocabulary matching against the CAP library, so only the organ and diagnosis strings change, never the reasoning edges.

3.6 Training

The Virchow2 encoder is frozen; only the MIL, pooler, and answer heads are trained on the precomputed bags. The organ head is supervised by cross-entropy

and each answer node by a focal loss [7] with $\gamma=1.5$ over its candidate set, combined as

$$\mathcal{L} = \lambda_o \text{CE}(\mathbf{p}, y_{\text{org}}) + \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \text{FL}_{\gamma}(s_q, y_q), \quad (5)$$

with $\lambda_o=8$ to protect the router that gates the entire walk. Answer heads are conditioned on the detached organ posterior, so gradients do not leak organ labels. We optimize with AdamW (learning rate 2×10^{-4} , weight decay 10^{-4}), a 5% linear warmup then cosine decay, gradient accumulation of 8, and gradient-norm clipping at 5, for 12 epochs. The split is patient-grouped and stratified by (organ, primary diagnosis) at 80/10/10; we keep the checkpoint with the best validation composite, the mean of macro organ accuracy and answer accuracy.

3.7 Inference

Given a WSI, GrandQC produces a tissue mask, the slide is tiled at 20x, and the frozen encoder yields the bag H ; the MIL head predicts the organ and, with $D(\bar{\mathbf{h}})$, decides in- versus out-of-distribution routing. From the organ root the walker repeats: pool the evidence $\mathbf{e}_q$, select $\hat{a}$, and emit $(q, \hat{a}, \text{next})$, where the next question(s) are the mined successor set $T(\text{organ}, q, \hat{a})$ for that context. Every emitted edge is thus a transition seen in the expert data; strings are canonicalized to the evaluation convention, so an edge is correct whenever it names the same transition. The walk terminates at the reporting node.

3.8 Report rendering

When the walk reaches the reporting node, the completed chain is rendered into a structured report with an organ-specific template that fills synoptic fields (histologic type, grade, extent, and grade detail such as the Gleason score or Nottingham components) from the answers already produced. Because the report is a function of the reasoning chain, each field traces to a produced answer and the report is faithful by construction: it cannot state a finding the reasoning did not reach.

3.9 Visual grounding responder

The visual-grounding interface receives a region of interest. A tissue-versus-background gate selects between two deterministic responses: a refusal on non-diagnostic background and a morphology statement on tissue. Determinism makes the answer stable under perturbation, and the explicit refusal separates background from tissue.

4 Results

4.1 Data and evaluation protocol

The training data comprise 11,220 expert reasoning chains paired with whole-slide images from 10,496 patients across seven organs (prostate, breast, colon,

stomach, bladder, lung, cervix) and five sites. Both axes are imbalanced: organ counts range from breast (2,213 slides) to cervix (812), and one site contributes 76% of slides, so institution shift is present within training itself. The organ list is not exhaustive of the anatomy seen at evaluation, motivating the distribution-shift design. We split at the patient level, stratified by (organ, primary diagnosis) at 80/10/10: 8,400 training, 1,038 validation, and 1,058 held-out cases. Mining yields 92 distinct questions and 1,402 transition triples; for 95.0% of nodes the set of next questions is identical across cases, and a walker that emits the mined successor set attains a reasoning-edge F1 ceiling of 0.988 given correct answers. Headline results are the blind evaluation on the independent challenge test set; the component analysis uses the patient-grouped development split of 1,058 cases.

All trained components, namely the MIL head, the question-conditioned pooler, the answer heads, and the distribution-shift detector, are fit only on the challenge training data. Virchow2, CONCH, and KEEP are publicly released pretrained models and the CAP protocol library is a public reference vocabulary, all used frozen at inference. Public TCGA slides from organs outside the seven are used solely to measure out-of-distribution generalization and never to fit or select any parameter.

We report every metric and state what it measures. *Binary path validity* is one only if the produced set of reasoning transitions exactly equals the reference set (all-or-nothing; a minor five-percent term, since our successor-set walker deliberately emits a superset). *Reasoning-edge* precision, recall, and F1 measure the overlap of the produced and reference transition sets, a transition being a (question, next-question) pair. *Matched-edge answer similarity* is the PubMedBERT [3] cosine between produced and reference answers on correctly recovered edges. The *report score* combines lexical overlap (ROUGE-L [6], BLEU-4 [12]), clinical-entity overlap (a scispaCy [10] named-entity Jaccard), and PubMedBERT similarity. *Visual grounding* scores correct refusal on non-tissue regions, answer consistency under perturbation, and tissue-versus-background discrimination, judged by a large language model (Qwen3 [20]).

4.2 Main results

Table 1 reports GOW on the blind Test Phase 1 evaluation, with an overall score of 0.791. Visual grounding is perfect (1.00 on background rejection, input sensitivity, and cross-region consistency): the deterministic tissue-versus-background gate stays stable under perturbation and across regions. The reasoning graph transfers well (edge F1 0.852), while the free-text report is the most shift-sensitive term (0.572), and binary path validity (0.483) is all-or-nothing and low by design, as the successor-set walker emits a superset of any single reference path (Section 5). On the development split held out from the training data the corresponding reasoning metrics are edge F1 0.922, matched-edge similarity 0.902, report 0.786, and binary path validity 0.600, with organ recognition (which gates the walk) 0.993.

Table 1. GOW on the blind Test Phase 1 evaluation. Visual grounding and its three robustness components are perfect. Binary path validity is all-or-nothing and low by design, since the successor-set walker emits a superset of any single reference path.

Metric	Test Phase 1
Overall score	0.791
Visual grounding	1.000
Background rejection	1.000
Input sensitivity	1.000
Cross-region consistency	1.000
Reasoning-edge F1	0.852
Matched-edge answer similarity (MESS)	0.772
Report score	0.572
Binary path validity	0.483

Table 2. Component contribution on the held-out split (n=400), with the OOD module off (its effect is reported separately). Each row removes or replaces one component.

Configuration	Edge F1	Answer sim.	Report
Modal chain (predicted organ)	0.777	0.778	0.555
GOW (full)	0.920	0.901	0.780
GOW without report enrichment	0.920	0.901	0.730
GOW without co-root seeding	0.882	0.829	0.447
GOW without successor-set walk	0.694	0.534	0.732

4.3 Component analysis

Table 2 isolates each component on the development split. Replacing the trained answer heads with the per-node modal answer costs the most, dropping edge F1 to 0.777 and the report to 0.555. Synoptic report enrichment acts specifically on the report term (0.780 to 0.730), leaving the edge and answer metrics unchanged. Co-root seeding injects the always-asked root questions such as the procedure, whose answers feed several report fields, so removing it drops the report to 0.447. Emitting a single successor instead of the mined set collapses edge F1 (0.920 to 0.694) and answer similarity (0.901 to 0.534), the fan-out the walker is built around. The distribution-shift module is off here (the split is in-distribution) and is evaluated separately below.

4.4 Distribution-shift detection

We validate the detector without any unseen-organ labels by leave-one-organ-out analysis: we fit on six organs, treat the seventh as unseen, and measure how well the distance separates the held-out organ from the six in-distribution organs, repeating for each organ. Replacing the shared covariance with per-organ covariances raises the averaged area under the ROC curve from 0.85 to 0.98, and

it is this quadratic form that lets a single detector guard all seven routes rather than one. We fix each per-organ threshold from in-distribution distances alone and scale it for precision, so that only about 2% of held-out in-distribution slides are flagged.

To test whether the detector generalizes beyond the training anatomy, we run it on public slides from organs outside the seven. It flags the dominant expected shift, uterine tissue routed to the cervix path, on every tested slide, and flags 9 of 10 further novel organs (brain, kidney, thyroid, liver, pancreas, skin, ovary, head and neck, adrenal); the one miss, esophagus, is routed to the adjacent gastric path. These slides are used only to measure generalization and never to fit any parameter. Routing is edge-safe: a flagged slide already follows its routed topology, so only the organ and diagnosis naming change, never the reasoning edges.

Naming the flagged organ is harder than detecting it: matching the shifted slide against the CAP vocabulary with two public encoders, CONCH and KEEP [21], recovers the organ for about four of ten novel organs open-vocabulary and six of ten within a constrained set. The residual error reflects genuine morphological overlap (clear-cell renal carcinoma versus adrenal cortex, squamous versus squamous) rather than the encoder, so detection is reliable and naming bounded.

5 Discussion and Limitations

Emitting the mined successor set maximizes edge coverage at the cost of exact path-validity, since the produced graph is a superset of any single reference path. GOW is bounded by a closed ontology: verifiable within that vocabulary, but unable to invent structure outside it. A few findings require magnification beyond 20x.

The detector responds to any departure from the training distribution, entangling organ novelty with scanner shift: under a leave-one-scanner-out analysis a Mahalanobis gate on the frozen bag-mean flags every held-out-scanner slide, trained organs included. Scoring instead in the router’s learned space with a scanner-covariance-aligned relative Mahalanobis cuts this false-flag rate to a few percent while still catching the uterine shift, at some cost to recall of subtler novel organs; institution-versus-organ disentanglement remains open. The shipped gate is calibrated for the challenge distribution, where organ shift dominates and the false-positive rate is two percent.

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